Topology Optimization
Lecture 11

Local versus global stress constraint strategies in topology optimization: A comparative study

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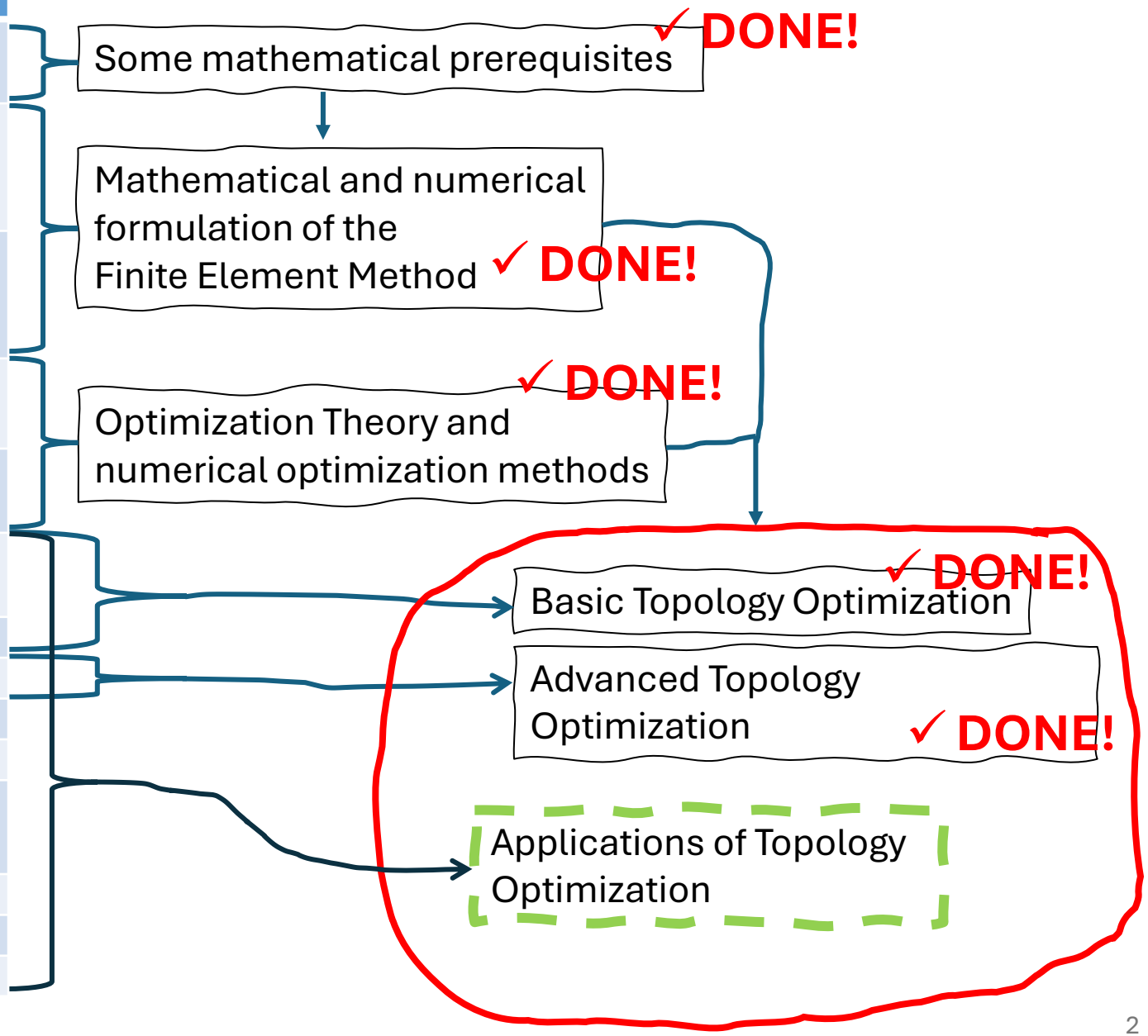
Int. J. for Numerical Methods in Engineering **122**, Issue 21, p. 6003-6036 (2021)
https://doi.org/10.1002/nme.6781

Course overview

No.	Topics
1	Introduction Function and vector spaces, Interpolation, L^2 projection
2	The Finite Element Method for PDEs 1: Poisson's Equation, The Finite Element Method, Properties of the Finite Element Method
3	The Finite Element Method for PDEs 2: Index notation, Linear Elasticity, Non-linear Poisson's equation, Non-linear Elasticity
4	Constrained Optimization 1: Optimality, KKT Theorem, Numerical Optimization Algorithms
5	Constrained Optimization 2: Method of Moving Asymptotes, MPI and PETSc
6	Introduction to Topology Optimization: The idea, Filtering and Projection, Python implementation
7	Gradients
8	Robust topology optimization, More on MPI
9	Wave equations
10	Acoustic waves and more on Robust Optimization
11	Penalty and Augmented Lagrangian Methods Stress-constraints in Topology Optimization 1
12	Stress-constraints in Topology Optimization 2
13	Light concentration using nanoparticle design
14	Heat conduction



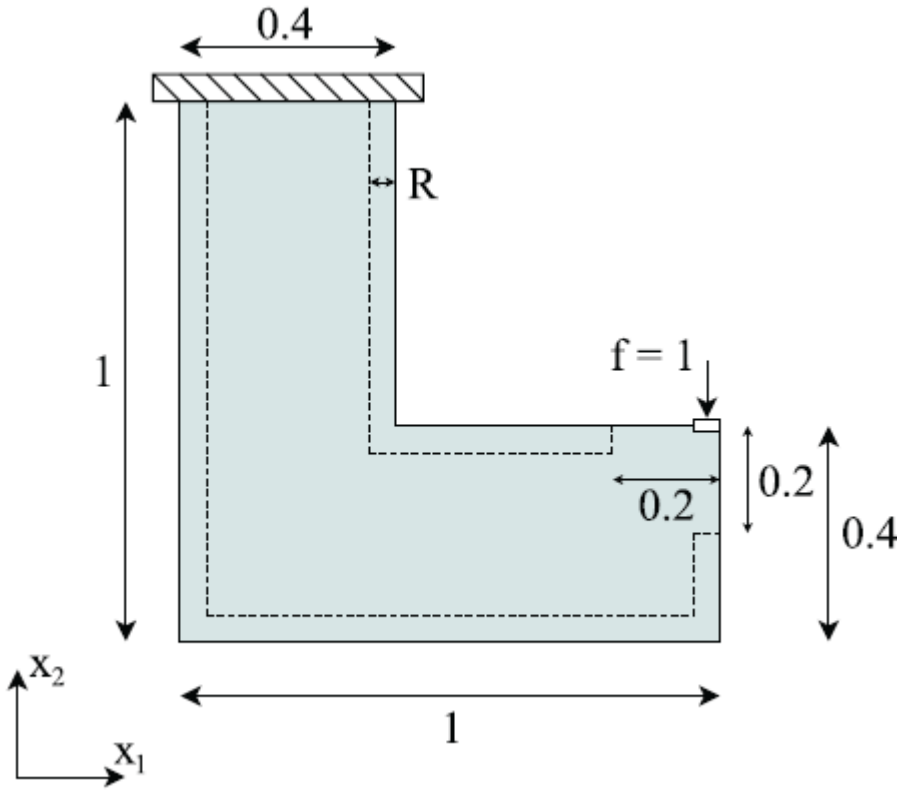
Stress constraints

- Stress is very important to keep under control to avoid failure of a mechanical component.
- Compliance-based optimization do not look at the stress, so it will have to be considered in "post analysis" of the design.
- In a continuous setting, the stress constraint can be written as
  - $\sigma_m(\boldsymbol{x}) \leq \sigma_y \; \forall \boldsymbol{x} \in \Omega$
  - $\sigma_m(\boldsymbol{x})$ is some stress measure at point $\boldsymbol{x}$, for example the von Mises stress.
  - σ_y is the yield or failure stress of the material.
- The constraint is often re-formulated to
  - $\frac{\sigma_m(\boldsymbol{x})}{f_\alpha(\sigma_y)} - 1 \leq 0 \; \forall \boldsymbol{x} \in \Omega$
  - $f_\alpha(\sigma_y)$ is a function of the yield stress, e.g., $f_\alpha(\sigma_y) = \alpha\sigma_y$ with $\alpha \leq 1$ acting as a 'safety factor'.
- Including stress-constraints in the optimization problem have several challenges:
  - The stress should be considered locally, usually 'in each mesh element'.
  - How to handle the stress singularity phenomena.
  - The stress can be artificially concentrated during the design process.
- Stress constraints are thus quite difficult to accurately take into account.

Many constraints

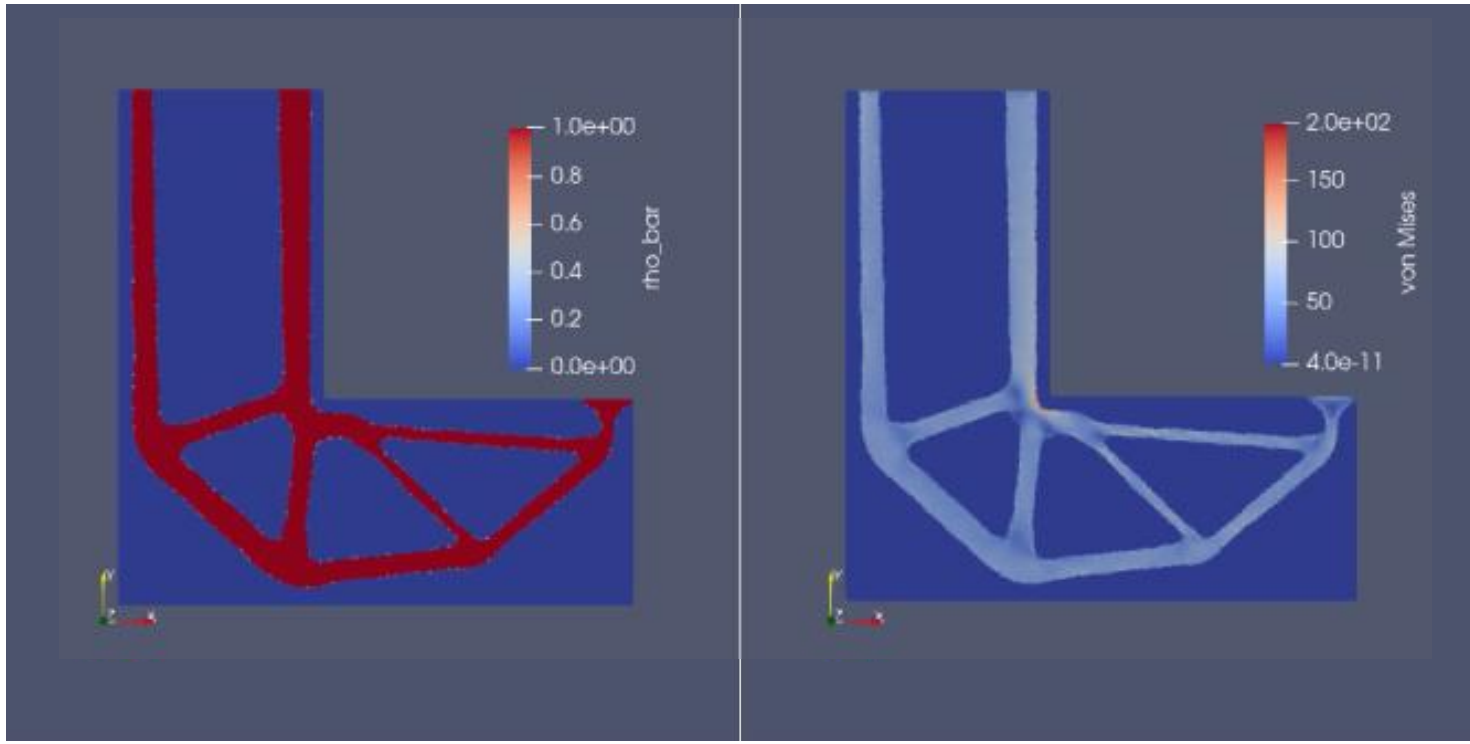
- With one constraint per mesh element, you have often >10.000 inequality constraints.
  - MMA would choke if we gave it an optimization problem with >10.000 constraints.
- We will consider two strategies:
  - Replace the many constraints with ONE global constraint.
  - Use a different solution strategy that can handle the many constraints.
- We will look at both approaches today and next time, starting with one global constraint today.
- “One global constraint” is the easiest to implement…

L-shaped problem



The main problem considered in the paper.
A 'padding' region with $\rho = 0$ is used outside the dashed lines.
Plane stress is used.

Double-filtered robust maximum stiffness optimization with volume constraint $V_f \leq 0.25$.



Calculation done 'as usual'.
Maximum von Mises stress is 200 'units', concentrated in the reentrant corner.

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The stress singularity phenomena

- The strain in low-density regions can be very high.
- When the density goes to 0 and *E* to nearly 0, the stress can tend to a non-zero value.
- The remaining stress in a region with 0 density can be even higher than the yield stress!
- When the density becomes 0 in a region, one should also remove the stress constraint for that region.
- The stress constrain function is thus discontinuous!
- One can not solve the optimization problem with ‘normal’ algorithms.

Cheng, G.D., Guo, X. ε-relaxed approach in structural topology optimization. *Structural Optimization* **13**, 258–266 (1997). https://doi.org/10.1007/BF01197454
P. Duysinx, O. Sigmund, New developments in handling stress constraints in optimal material distribution, DOI: 10.2514/6.1998-4906, Conference: 7th AIAA/USAF/NASA/ISSMO Symposium on Multidisciplinary Analysis and Optimization

The stress singularity phenomena

- A solution to the problem is to increase the yield stress for low density elements, e.g.,

$$\sigma_{vM} \leq \sigma_y \left(1 - \epsilon + \frac{\epsilon}{\rho}\right)$$

  and let $\epsilon \to 0$ in a continuation scheme.
  - For $\rho = 1$, the r.h.s. give σ_y.
  - When $\rho \to 0$, the r.h.s. diverges, effectively removing the constraint, unless $\epsilon = 0$.
- This is known as ϵ-relaxation.
- Today's paper formulates the stress constraints as

$$f_\sigma(\bar{\rho})\sqrt{\sigma_{vM}^2 + \sigma_{\min}^2} \leq \sigma_y \text{ with}$$

$$f_\sigma(\bar{\rho}) = \frac{1}{1 - \epsilon + \frac{\epsilon}{\bar{\rho}}} = \frac{\bar{\rho}}{\epsilon(1 - \bar{\rho}) + \bar{\rho}}, \qquad \sigma_{vM}^2 = \frac{3}{2} s_{ij} s_{ij}, \qquad s_{ij} = \sigma_{ij} - \frac{1}{3}\sigma_{kk}\delta_{ij}$$

- $\sigma_{\min} = 10^{-4}\sigma_y$ is introduced in the paper to ensure better numerical stability as $\sigma_{vM}^2 \to 0$.
- σ_{vM} is calculated with the elastic modulus of the base material, not the “SIMP” elastic modulus.
  - Alternatively, $\sigma_{vM}/\bar{\rho}^\eta$ can be used, where η is the exponent used in the SIMP interpolation (often $\eta = 3$). Other relaxation methods exist to solve the problem with low density elements.
  - qp-relaxation is another popular choice.

P. Duysinx, M. P. Bendsøe, Topology optimization of continuum structures with local stress constraints, Int. J. Numer. Meth. Engng. **43**, 1453 (1998)
Cheng, G.D., Guo, X. ε-relaxed approach in structural topology optimization. *Structural Optimization* **13**, 258–266 (1997). https://doi.org/10.1007/BF01197454
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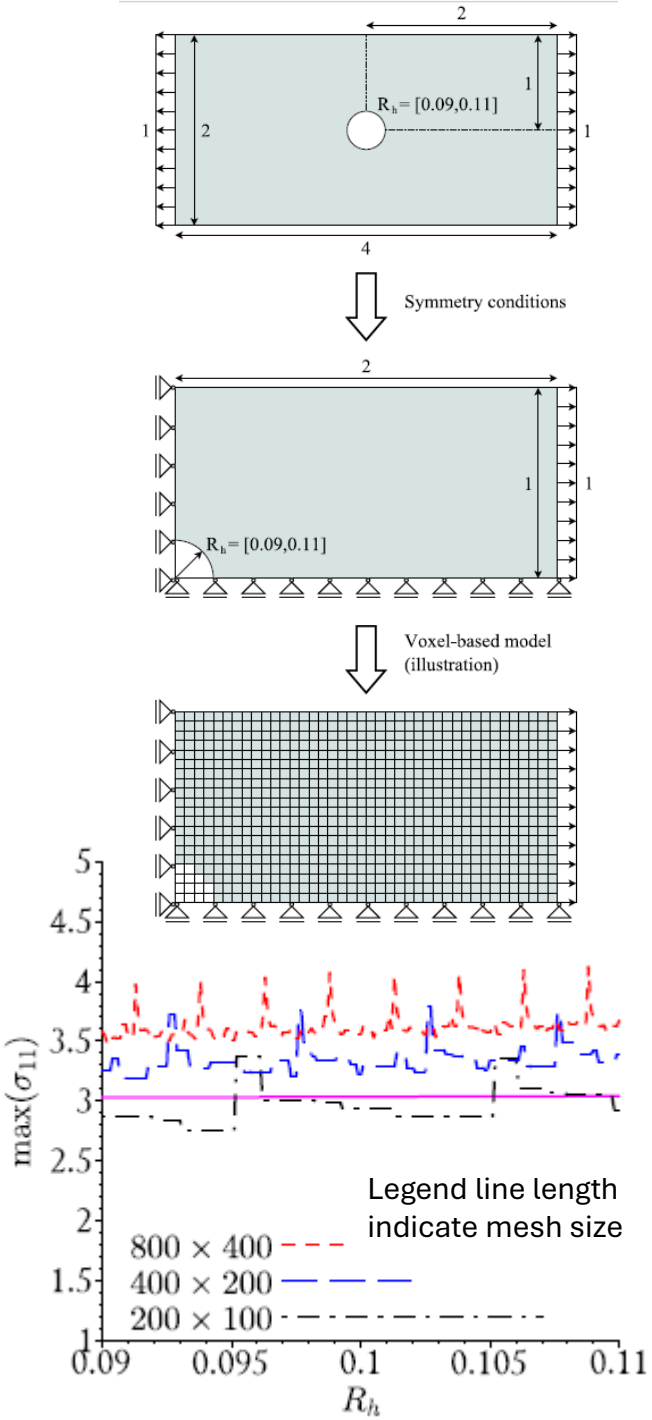
Artificial stress concentration



- The pixel/voxel-based geometry used in topology optimization do not represent the stress at the 'jagged' boundaries very well!
- da Silva et. al. considered a plate with a hole and looked at $\max(\sigma_{11})$ and found big differences between the topology optimized 'fixed grid' 0-1 discretization and the 'normal' CAD discretization with a 'real' hole.
- The differences got larger when the pixel size was made smaller, so a finer mesh in topology optimization does not help.
- Plotting the stress for different hole radii, R_h, a period pattern appear. The periodicity is roughly the mesh size.
- The authors demonstrate that a simple approach to limit this artificial stress concentration at jagged edges is to allow a small soft layer of intermediate density between the solid and void phases.

G.A. da Silva, A.T. Beck and O. Sigmund, Computer Methods in Applied Mechanics and Engineering 344 (2019) 512–537

Artificial stress concentration

- For a convolution filter, da Silva et. al. demonstrate that the soft layer is obtained by limiting the maximum value of the projection strength β to $\beta_{lim} = \frac{2R}{l_e}$, where R is the convolution filter radius and l_e is the mesh size.
- For a PDE filter $\beta_{lim} = \frac{2R}{l_e\sqrt{3}} = \frac{4R_{PDE}}{l_e}$ since $R_{PDE} = \frac{R}{2\sqrt{3}}$ where R_{PDE} is the parameter used in the Helmholtz equation.
- da Silva et. al. considered also the connection between the ϵ relaxation parameter and the maximum value of β used.
- Good agreement between the maximum stress for the pixel discretization and the real hole problem was found when using $\beta \leq \frac{\beta_{lim}}{2}$ and fixed $\epsilon = 0.2$.
- The study is based on using SIMP interpolation with penalization exponent 3.
  - If another interpolation is used, a new study needs to be made to find the best parameters for accurate stress calculation.
- It is highly advisable to always check the stress level of the optimized design in a "real" FEM calculation with smooth/CAD geometry and converged mesh size.

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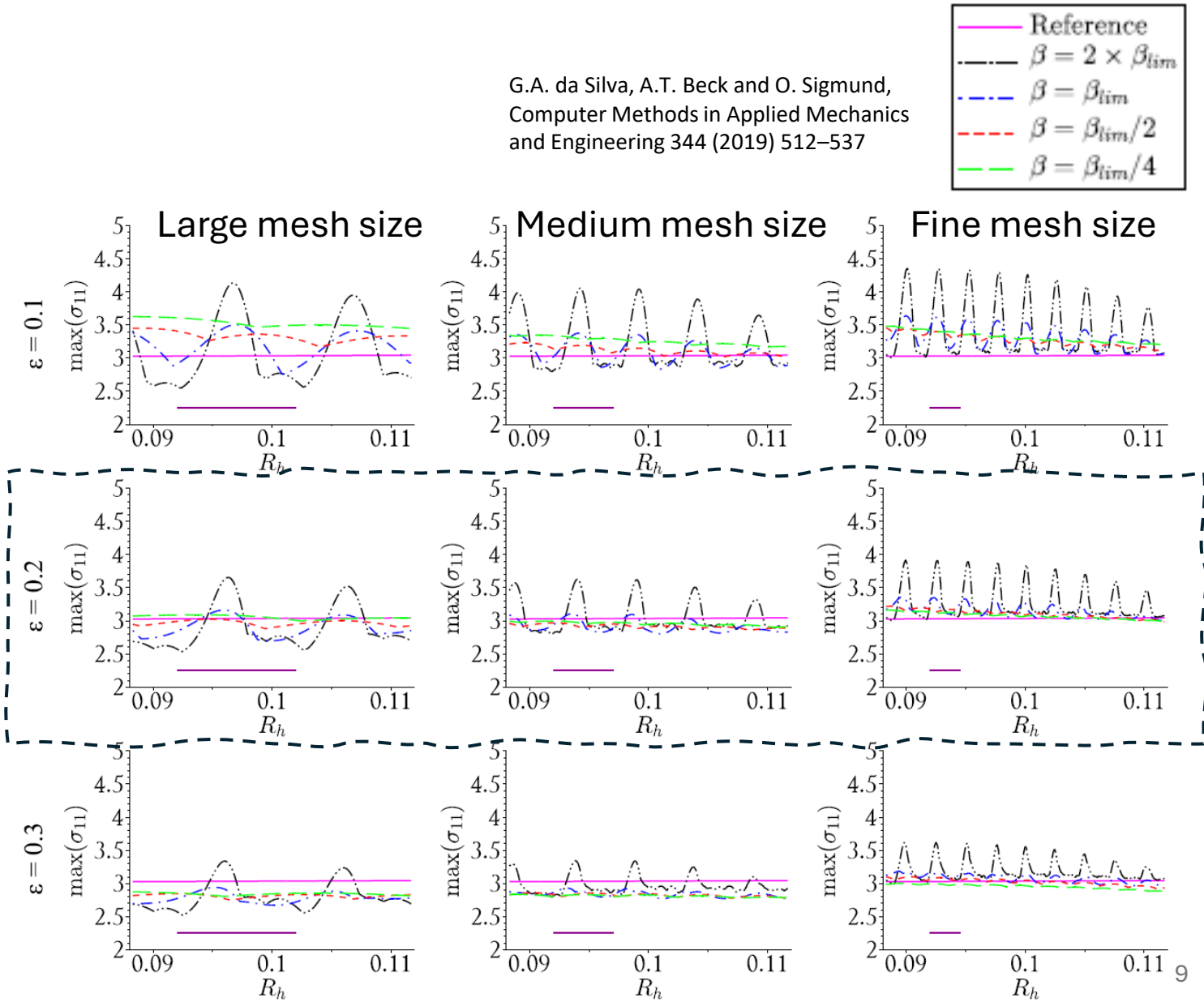


One global constraint

- Replace the many constraints by one global constraint

$$\sigma_m(\boldsymbol{x}) \leq f_\alpha(\sigma_y)\ \forall \boldsymbol{x} \in \Omega \quad \Rightarrow \quad \left(\max_{\boldsymbol{x}} g(\boldsymbol{x})\right) - 1 \leq 0 \quad \text{with} \quad g(\boldsymbol{x}) = \frac{\sigma_m(\boldsymbol{x})}{f_\alpha(\sigma_y)}$$

- $\max$ is not a differentiable function, so gradients cannot be calculated ☹
- Replace $\max$ by a differentiable approximation, e.g.,
  - The p-norm: $\max_{\boldsymbol{x}} g(\boldsymbol{x}) \approx \max_i g_i \approx \left(\sum_{i=1}^N g_i^p\right)^{1/p} \approx \left(\int_\Omega g^p(\boldsymbol{x}) d\Omega\right)^{1/p}$
  - The p-mean: $\max_{\boldsymbol{x}} g(\boldsymbol{x}) \approx \max_i g_i \approx \left(\frac{1}{N}\sum_{i=1}^N g_i^p\right)^{1/p} \approx \left(\frac{1}{|\Omega|}\int_\Omega g^p(\boldsymbol{x}) d\Omega\right)^{1/p}$
  - g_i^p is the function measured in mesh element number i raised to power p.
- Both the p-norm and the p-mean tends to the $\max$ as $p \to \infty$:
  - The discrete p-norm values are higher than the max for finite p.
  - The continuous p-norm and p-mean values are lower than the max for finite p.
- Today's paper chooses to use the p-mean.
- In FEniCS, we don't use sums but use instead the integral approximations.

One global constraint

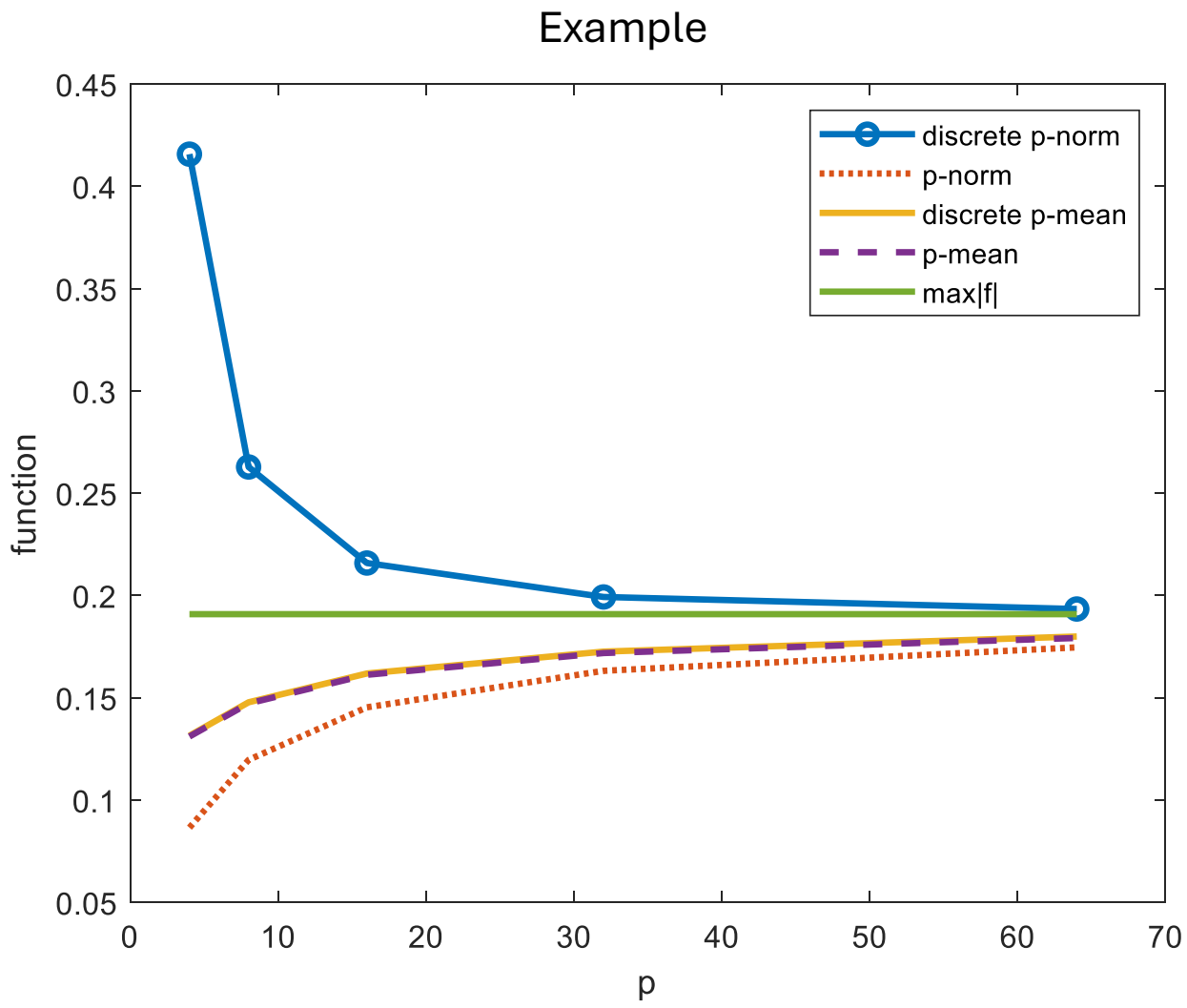
$$\max_{\boldsymbol{x}} g(\boldsymbol{x})$$

$$\approx \left(\sum_{i=1}^{N} g_i^p\right)^{\frac{1}{p}}$$ Discrete p-norm.

$$\approx \left(\int_{\Omega} g^p(\boldsymbol{x}) d\Omega\right)^{1/p}$$ p-norm.

$$\approx \left(\frac{1}{N}\sum_{i=1}^{N} g_i^p\right)^{\frac{1}{p}}$$ Discrete p-mean.

$$\approx \left(\frac{1}{|\Omega|}\int_{\Omega} g^p(\boldsymbol{x}) d\Omega\right)^{1/p}$$ p-mean.



One global constraint

- The stress constraint

$$\sigma_m(\boldsymbol{x}) \leq f_\alpha(\sigma_y)\ \forall \boldsymbol{x} \in \Omega \quad \Rightarrow \quad \max_{\boldsymbol{x}} g(\boldsymbol{x}) - 1 \leq 0 \quad \text{with} \quad g(\boldsymbol{x}) = \frac{\sigma_m(\boldsymbol{x})}{f_\alpha(\sigma_y)}$$

  is approximated by

$$\left(\frac{1}{|\Omega|}\int_\Omega \left(\frac{\sigma_m(\boldsymbol{x})}{f_\alpha(\sigma_y)}\right)^p d\Omega\right)^{\frac{1}{p}} - 1 \leq 0 \quad \Rightarrow \quad \sigma_c - 1 \leq 0 \quad \text{where} \quad \sigma_c = \left(\frac{1}{|\Omega|}\int_\Omega \left(\frac{\sigma_m(\boldsymbol{x})}{f_\alpha(\sigma_y)}\right)^p d\Omega\right)^{\frac{1}{p}}$$

- Since the *p*-mean underestimates the stress for low *p* values and the optimization problem is very difficult to solve for high *p*-values, some strategies can be followed:
  - Adaptive normalization: Adjust the constraint every iteration to take the underestimation into account:

$$c^b \sigma_c^b - 1 \leq 0, \qquad c^b = q\frac{\max\left(\sigma_m\left(\bar{\rho}^{b-1}\right)\right)}{f_\alpha(\sigma_y)\sigma_c^{b-1}} + (1-q)c^{b-1}, c^0 = 1,$$

    - q is a parameter, which the paper set to $q = \frac{1}{2}$.
    - $q = 0$ turn off the adaptive normalization.
    - *b* counts the iterations of the optimization solver, e.g., outer MMA iterations.
    - One can also change c^b only every N^{th} iteration instead of every iteration (N=1).
  - Increase *p* in a continuation scheme to be able to reach very high *p*-values, e.g., *p*>100.
  - A combination of the above two strategies.

One global constraint

- The stress constraints are thus implemented by having the following "global" constraint in the topology optimization problem

$$\max_{\boldsymbol{x}}\left(\sigma_m(\boldsymbol{x})\right) \leq f_\alpha\left(\sigma_y\right)$$

approximated as

$$c^b \sigma_c^b - 1 \leq 0 \text{ with}$$

$$\sigma_c = \left(\int_\Omega \frac{1}{|\Omega|}\left(\frac{\sigma_m}{f_\alpha\left(\sigma_y\right)}\right)^p d\Omega\right)^{\frac{1}{p}}$$

$$c^b = q\frac{\max\left(\sigma_m\left(\bar{\rho}^{b-1}\right)\right)}{f_\alpha\left(\sigma_y\right)\sigma_c^{b-1}} + (1-q)c^{b-1}, \quad c^0 = 1, \quad f_\alpha\left(\sigma_y\right) = \alpha\sigma_y$$

$$\sigma_m(\boldsymbol{x}) = f_\sigma(\bar{\rho})\sqrt{\sigma_{vM}^2 + \sigma_{\min}^2},$$

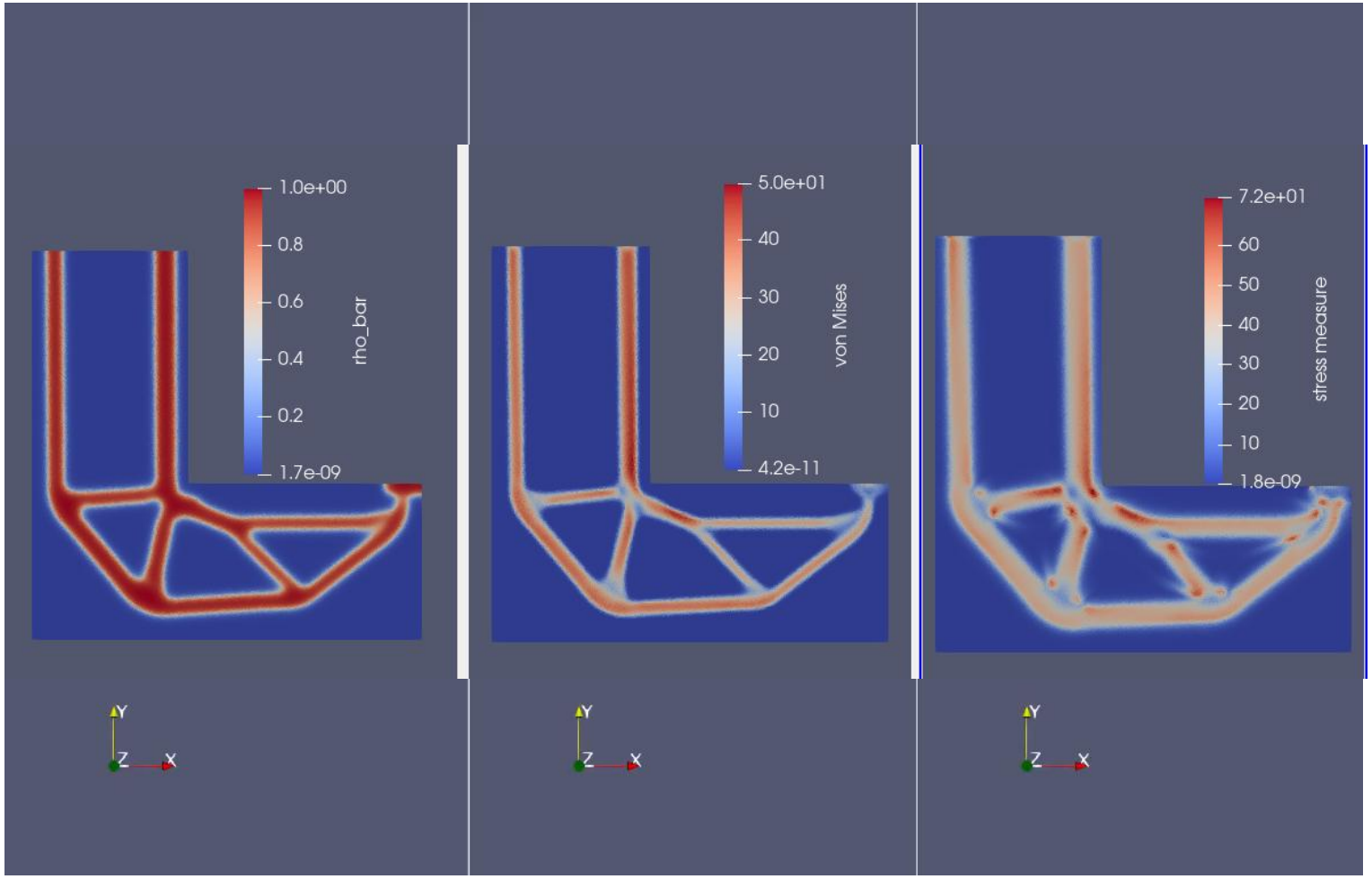
$$f_\sigma(\bar{\rho}) = \frac{\bar{\rho}}{\epsilon(1-\bar{\rho}) + \bar{\rho}},$$

$$\sigma_{vM}^2 = \frac{3}{2}s_{ij}s_{ij}, \; s_{ij} = \sigma_{ij} - \frac{1}{3}\sigma_{kk}\delta_{ij}$$

- b is the optimization solver iteration counter. c^b can also be update only every N^{th} iteration instead of every iteration (N=1).
- The implementation should have fixed $\epsilon = 0.2$ and SIMP penalization 3.
- σ_{vM} is calculated with the elastic modulus of the base material, not the "SIMP" elastic modulus.
- The paper uses $\sigma_{\min} = 10^{-4}\sigma_y$ and $q = \frac{1}{2}$.
- β and p should be raised in a continuation scheme, with $\beta \leq \beta_{lim}/2$ and "$p \to \infty$". Starting values used in the paper are $\beta = 0.1$ and $p = 2$.
- In a robust formulation, one stress constraint should be added per design realization (often 3).
- When implementing this, start without adaptive normalization (q =0) and implement adaptive normalization if required.

L-shaped problem

Double filtered robust optimization with global stress constraint, $\sigma_{vM} \leq 70$.



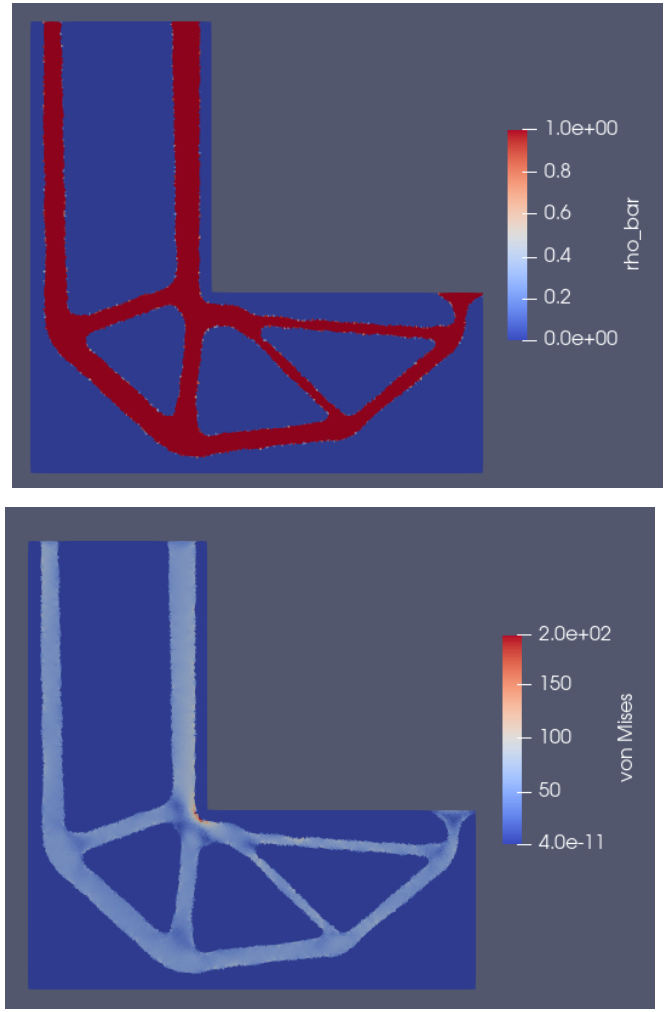
The cost function used here was $\int_\Omega \frac{1}{2}\sigma_{ij}\epsilon_{ij}d\Omega$ and volume constraint $V_f \leq 0.25$.

β, p, c^b were all updated after 20 MMA iterations.

β was raised to $\beta_{\text{lim}}/2$ and p was raised to 300 over 16 step.

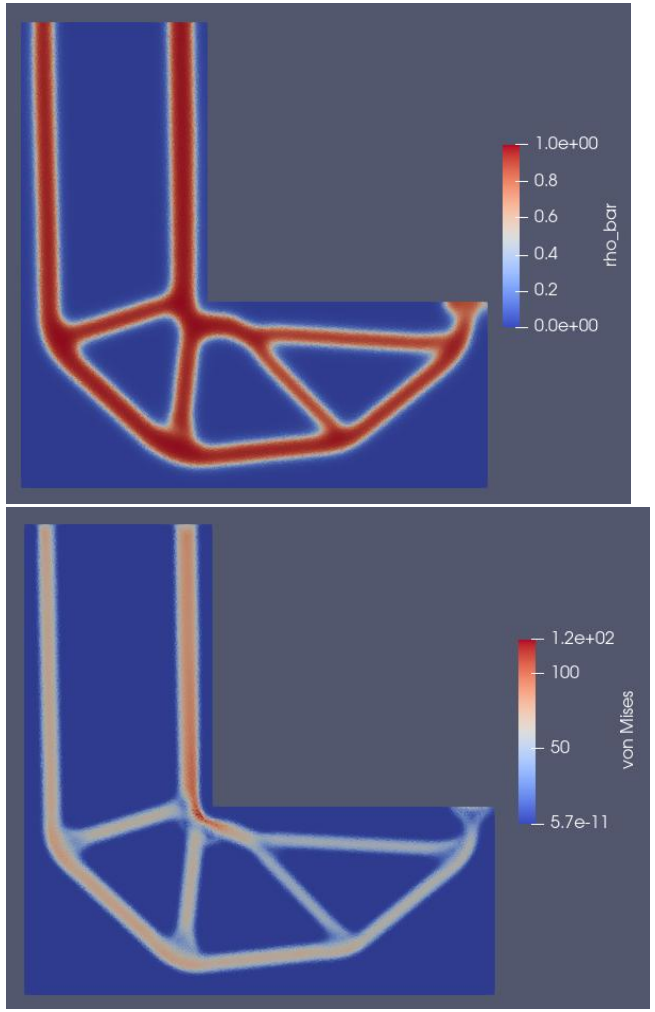
L-shaped problem

Compliance, $\beta_{max} = 64$



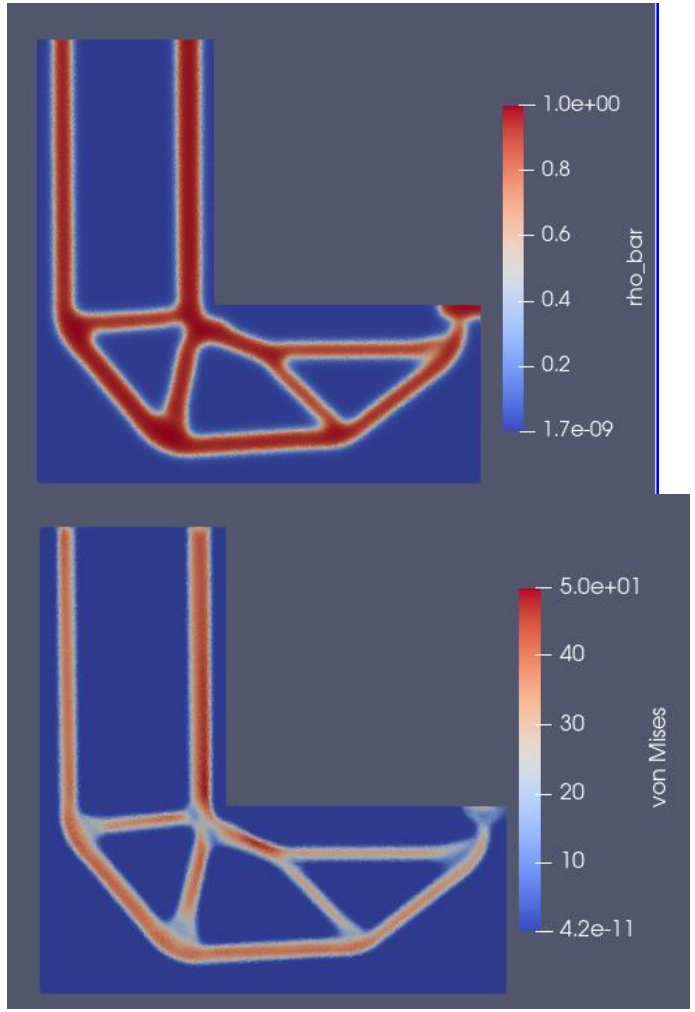
Max von Mises 200 ‘units’.

Compliance, $\beta_{max} = 3.5$



Max von Mises 120 ‘units’.

Stress constraints, $\beta_{max} = 3.5$



Max von Mises 50 ‘units’.

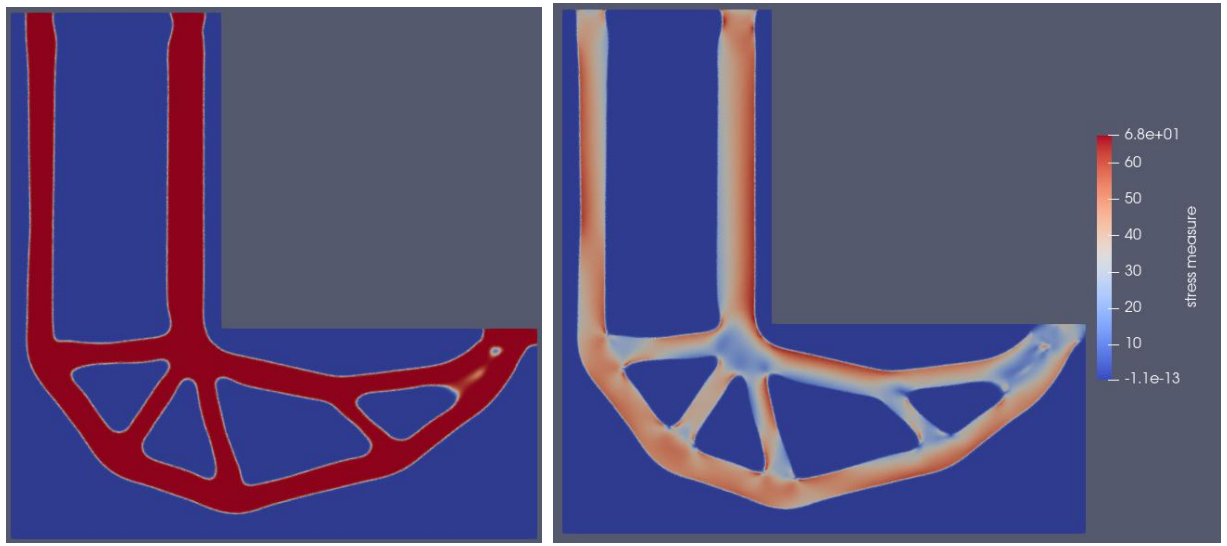
Note: The von Mises stress calculated in the topology optimization may be slightly incorrect. Stresses should (always) be re-calculated in post-processing of the final design using a ‘real’ FEM model with smooth surface and converged mesh to check the failure criteria!

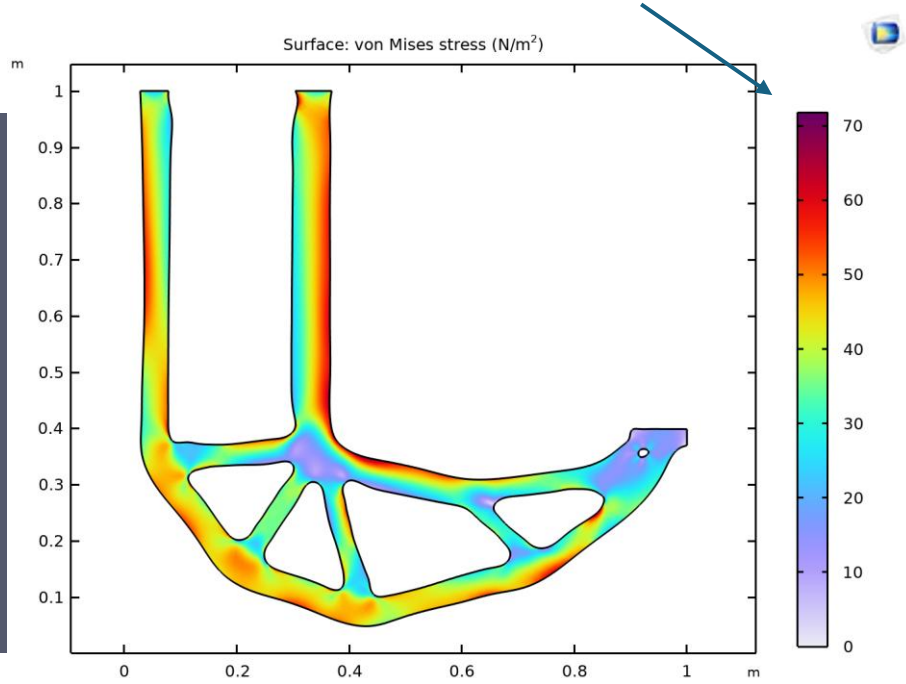
L-shaped example

Maximum only slightly > 70…

Optimized with

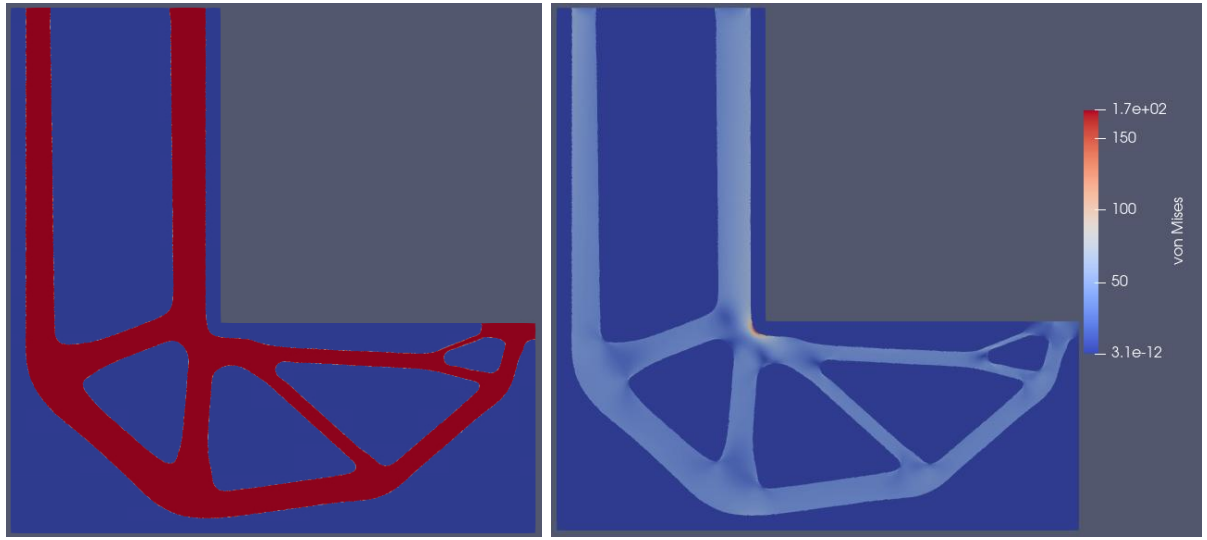
$$\min \int_\Omega \frac{1}{2}\sigma_{ij}^N \epsilon_{ij}^N d\Omega$$
$$\sigma_{vM}^k \leq 70$$
$$k \in \{\mathrm{E}, \mathrm{N}, \mathrm{D}\}$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 14$$
$$p_{\max} = 300$$

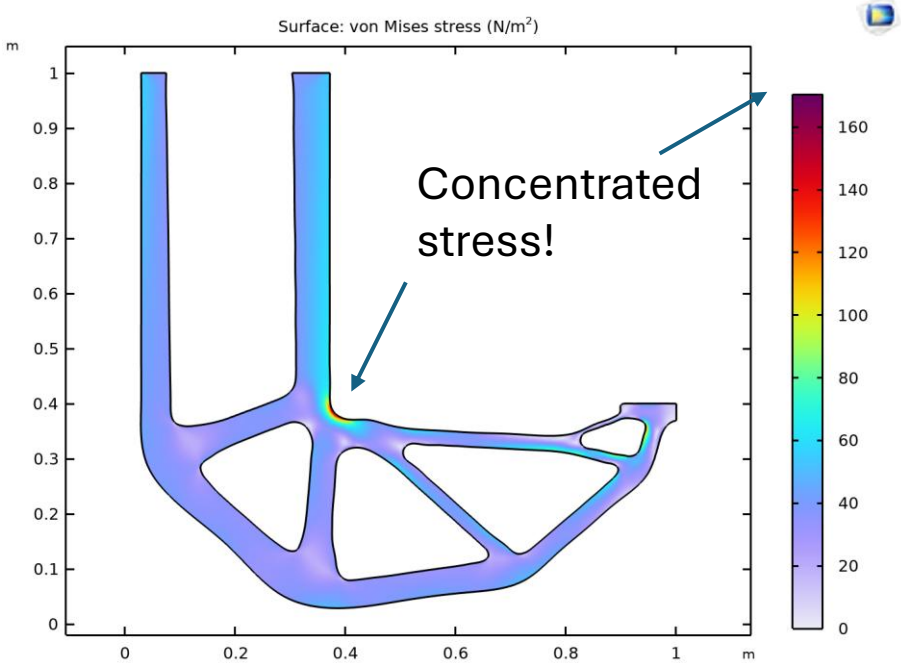




Optimized with

$$\min \max_{k \in \{\mathrm{E},\mathrm{N},\mathrm{D}\}} f^k$$
$$f^k = \int_\Omega \frac{1}{2}\sigma_{ij}^k \epsilon_{ij}^k d\Omega$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 64$$



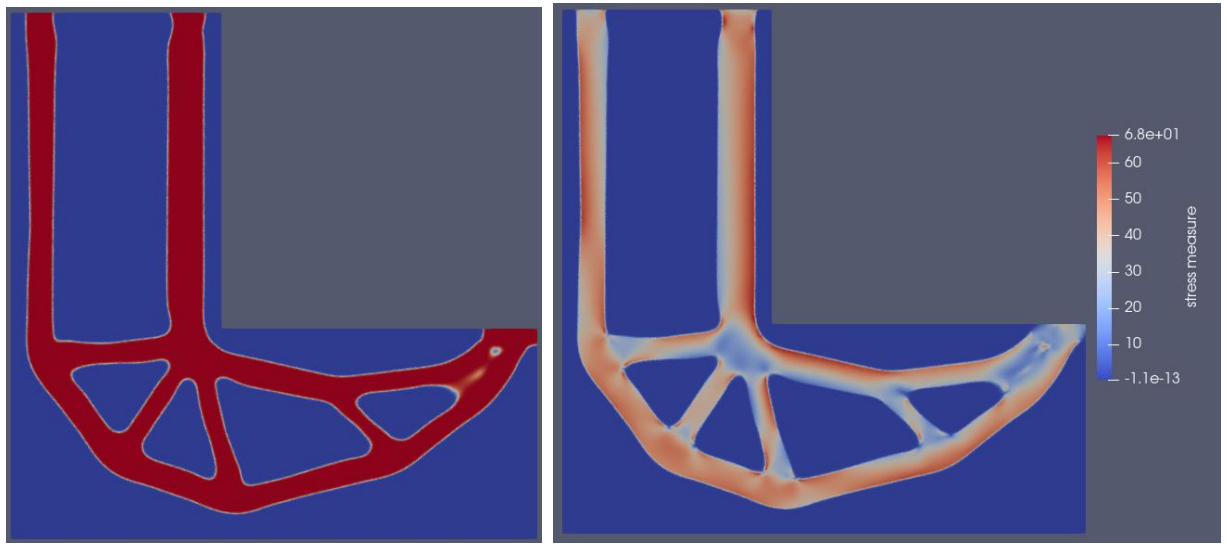


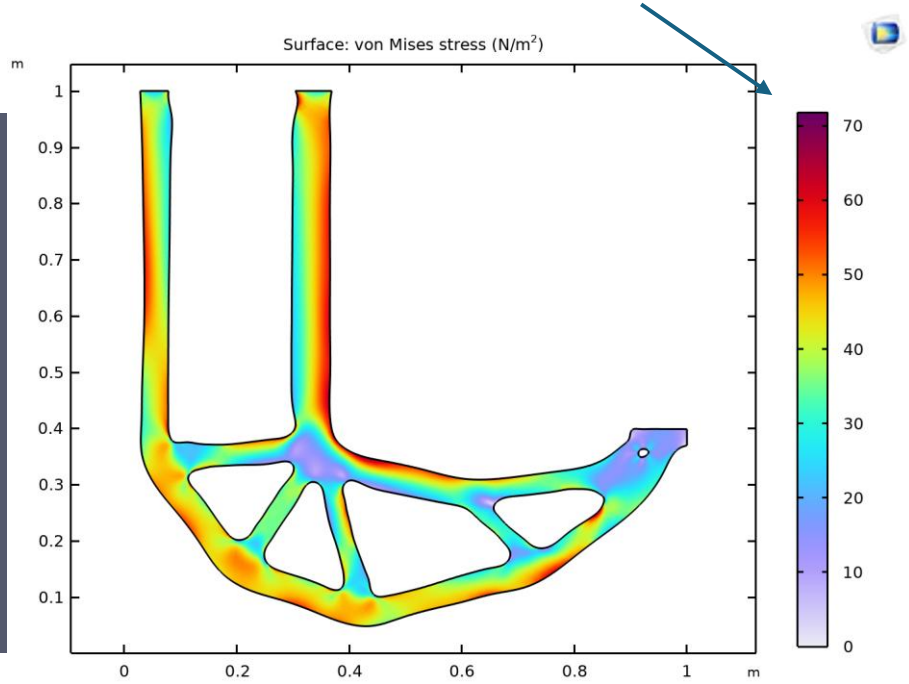
L-shaped example

Maximum only slightly > 70…

Optimized with

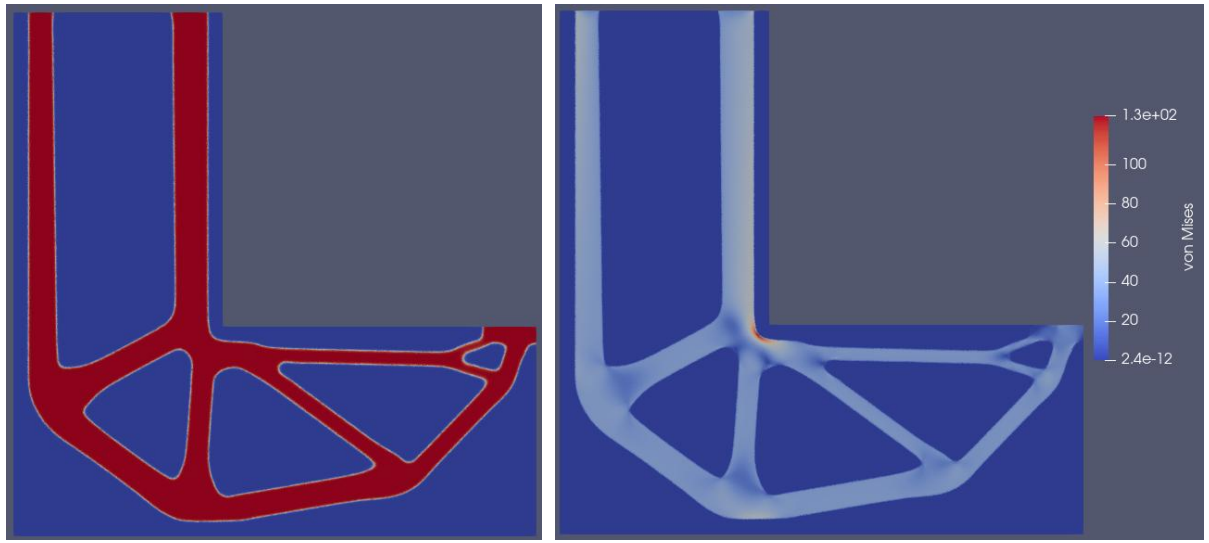
$$\min \int_\Omega \frac{1}{2}\sigma_{ij}^N \epsilon_{ij}^N d\Omega$$
$$\sigma_{vM}^k \leq 70$$
$$k \in \{\mathrm{E}, \mathrm{N}, \mathrm{D}\}$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 14$$
$$p_{\max} = 300$$

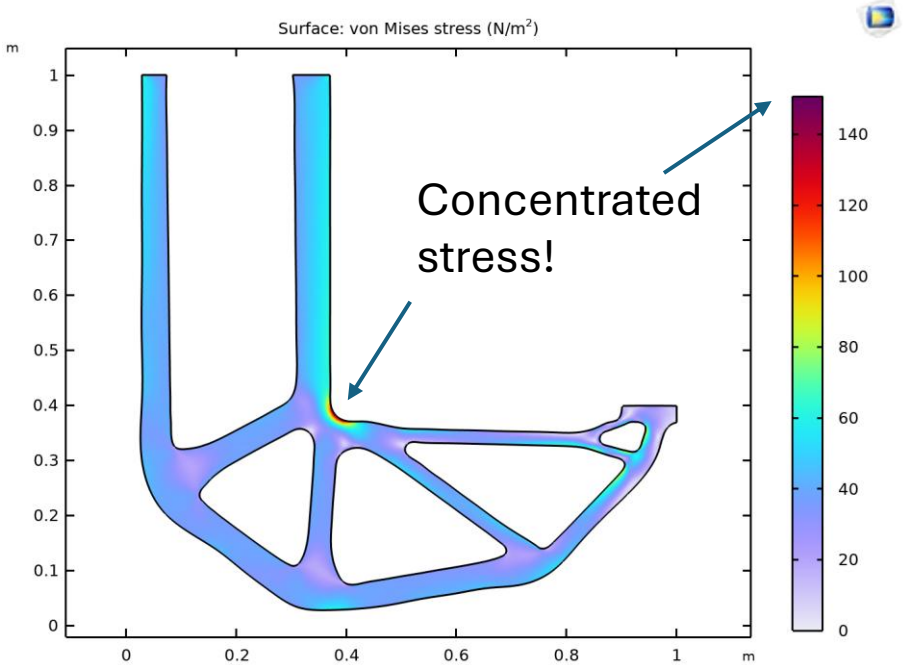




Optimized with

$$\min \int_\Omega \frac{1}{2}\sigma_{ij}^N \epsilon_{ij}^N d\Omega$$
$$\sigma_{vM}^k \leq 7000$$
$$k \in \{\mathrm{E}, \mathrm{N}, \mathrm{D}\}$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 14$$
$$p_{\max} = 3$$





L-shaped example

Optimized with

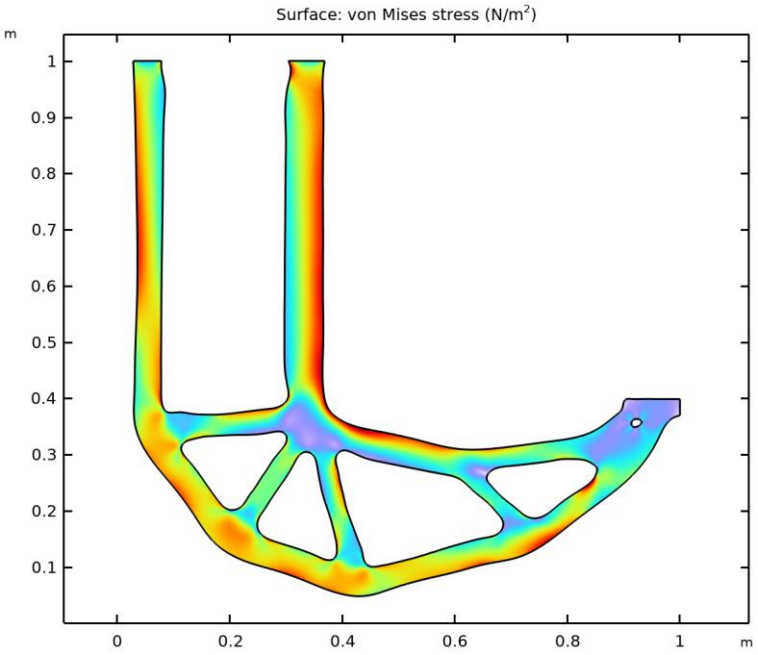
$$\min \int_\Omega \frac{1}{2}\sigma_{ij}\epsilon_{ij} d\Omega$$
$$\sigma_{vM}^k \leq 70\,, k \in \{\mathrm{E,N,D}\}$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 14$$
$$p_{\max} = 300$$

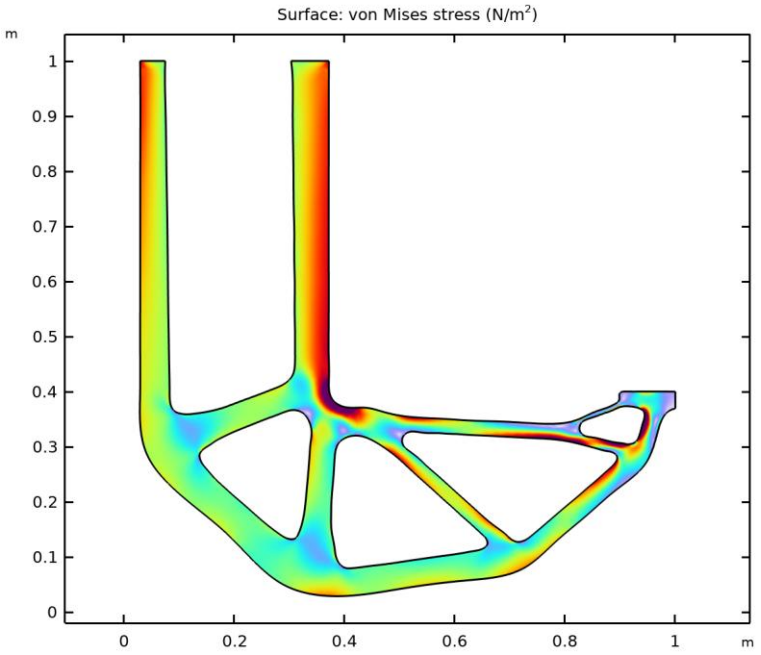
Optimized with

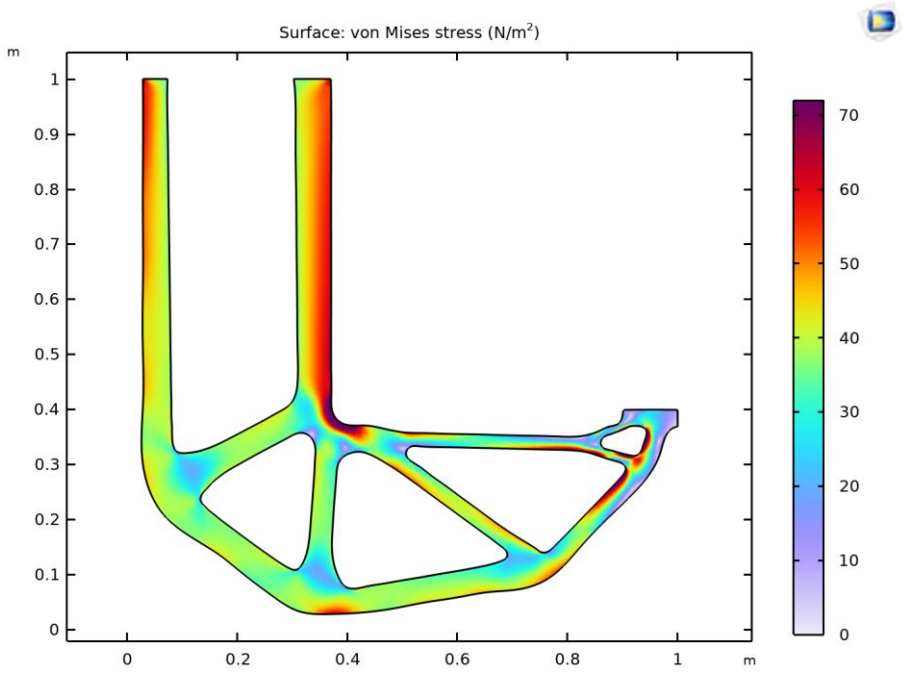
$$\min \max_{k\in\{\mathrm{E,N,D}\}} \int_\Omega \frac{1}{2}\sigma_{ij}^k\epsilon_{ij}^k d\Omega$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 64$$

Optimized with

$$\min \int_\Omega \frac{1}{2}\sigma_{ij}\epsilon_{ij} d\Omega$$
$$\sigma_{vM}^k \leq 7000\,, k \in \{\mathrm{E,N,D}\}$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 14$$
$$p_{\max} = 3$$







Many local constraints

- Consider an optimization problem with an infinite number of stress constraints

$$\min_{\rho} \mathrm{f}(\rho)\ s.t.\ \frac{\sigma_m\big(\rho(\boldsymbol{x})\big)}{f_\alpha\big(\sigma_y\big)} - 1 \leq 0\ \forall \boldsymbol{x} \in \Omega$$

  and write down the Lagrange function

$$L = f(\rho) + \int_\Omega \lambda(\boldsymbol{x}) \left(\frac{\sigma_m\big(\rho(\boldsymbol{x})\big)}{f_\alpha\big(\sigma_y\big)} - 1 \right) d\Omega$$

  where $\lambda(\boldsymbol{x})$ is a Lagrange multiplier field.

- There is not really any fundamental mathematical issues with an infinite number of constraints (or large finite number of constraints). It is only a problem for constrained optimization algorithms, as the constraints are considered explicitly in the solution process.
- Algorithms which aggregate all the constraints in a single function are able to handle the many constraints.
- Penalty methods and augmented Lagrangian methods do exactly that.

Nocedal and Wright

We will follow Nocedal and Wright for the Penalty method and the augmented Lagrangian method, so note the following:

Aurora, that we followed for studying the KKT theorem, define the Lagrange function (for the alternative form of the KKT conditions) as

$$L = f(\boldsymbol{x}) + \boldsymbol{v}^T \boldsymbol{h}(\boldsymbol{x}) + \boldsymbol{u}^T \boldsymbol{g}(\boldsymbol{x})$$

where $\boldsymbol{v}$ is a vector of Lagrange multipliers for equality constraints and $\boldsymbol{u}$ is a vector of Lagrange multipliers for inequality constraints.

Nocedal and Wright define the Lagrange function as

$$L = f(\boldsymbol{x}) - \boldsymbol{\lambda}^T \boldsymbol{c}(\boldsymbol{x})$$

where $\boldsymbol{\lambda}$ is a vector of Lagrange multipliers and $\boldsymbol{c}$ is a vector of all constraint functions, both equality and inequality.

The definitions are equivalent, but the signs of the Lagrange multipliers are opposite to Aurora and Aurora's notation show explicitly which constraints are equalities and which are inequalities.

Penalty Methods

To solve a constrained optimization problem using an unconstrained optimization algorithm, one can replace the original cost function by a new cost function which includes a penalty for violated constraints.

Consider the problem with equality constraints only:

$$\min f(\boldsymbol{x}) \text{ subject to } h_i(\boldsymbol{x}) = 0 \quad , \quad i = 1, ..., p$$

The new cost function is defined as:

$$Q(\boldsymbol{x}, \mu) = f(\boldsymbol{x}) + \frac{\mu}{2} \sum_{i=1}^{p} h_i^2(\boldsymbol{x})$$

This is termed the quardratic penalty function for the problem.

Now we do a series of unconstrained minimizations of Q for $\mu = \mu^1, \mu^2, ..., \mu^k$ where $\mu^{i+1} > \mu^i$ for $i = 1, ..., k-1$.

The solution to the $(i-1)$'th problem is used as the starting point for the i'th problem.

As $\mu^k \to \infty$, constraint violations give a larger and larger penalty to the new cost function, so to reduce these penalties the solution should show $h_i \to 0$ as $\mu^k \to \infty$.

Penalty Methods

The penalty method for equality constrained problems:

Choose some initial penalty factor μ^0 and tolerance ϵ^0.
Choose some overall tolerance, τ, and a maximum number of iterations, k_{max}.
Choose some starting point $\boldsymbol{x}^{-1}$.
Set $k = 0$.
While $k < k_{max}$ and ($k = 0$ or $||\boldsymbol{x}^{k-1} - \boldsymbol{x}^{k-2}|| < \tau$)
    Obtain $\boldsymbol{x}^k$ by
        minimizing $Q(\boldsymbol{x}, \mu^k) = f(\boldsymbol{x}) + \frac{\mu}{2} \sum_{i=1}^{p} h_i^2(\boldsymbol{x})$ wrt. $\boldsymbol{x}$
        from starting point $\boldsymbol{x}^{k-1}$ to tolerence ϵ^k.
    Choose $\mu^{k+1} > \mu^k$ and $\epsilon^{k+1} \leq \epsilon^k$.
    $k = k + 1$.

Most often, μ^k is increased by a constant factor, e.g. $\mu^{k+1} = q\mu^k$ with $q = 2$ or 10 or some other (for the specific problem) appropiate number.

One can also adapt the update of μ^k to how hard it is to solve the problem for the current value. If many iterations are used in the unconstrained minimization, make only a moderate update (e.g. $q = 1.5$). If few iterations are used, make a large update (e.g. $q = 10$).

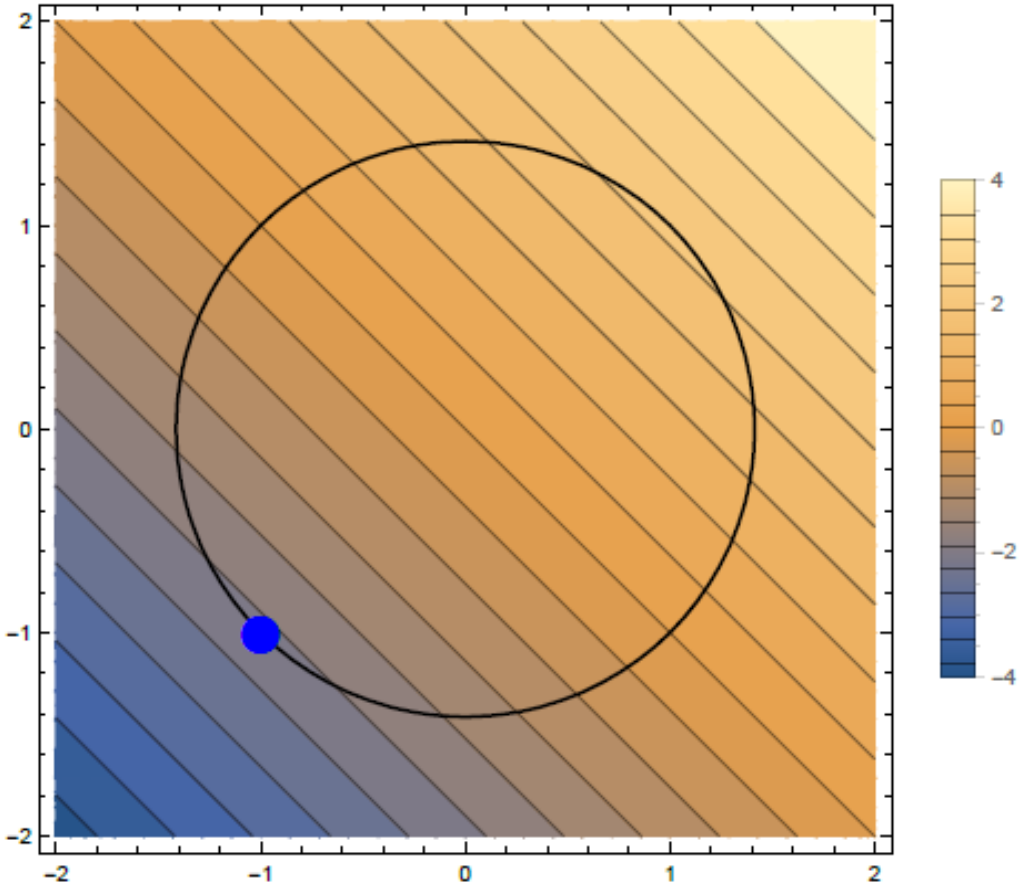
ϵ^k can be updated in a similar fashion, although a constant value for ϵ^k is used very often in practice.

Example

Consider the problem:

$$\min\ f(\boldsymbol{x}) = x_1 + x_2 \text{ subject to } h(\boldsymbol{x}) = x_1^2 + x_2^2 - 2 = 0$$

The constraint is a circle with radius $\sqrt{2}$ and the minimum is easily found analytically or graphically to be $x_1 = x_2 = -1$.



Example

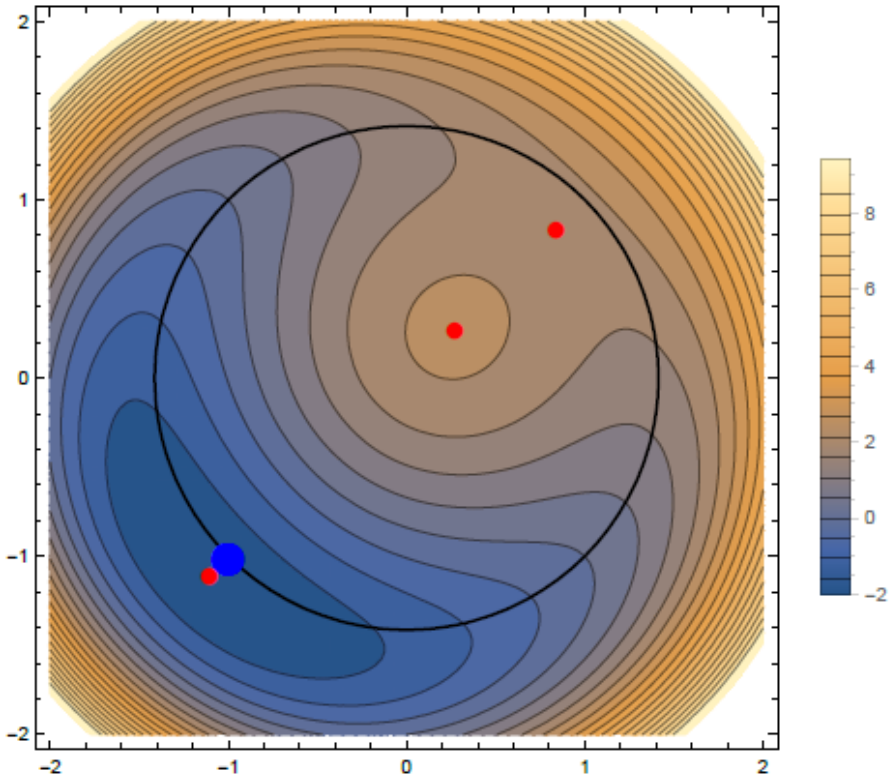
For the problem:

$$\min\ f(\boldsymbol{x}) = x_1 + x_2 \text{ subject to } h(\boldsymbol{x}) = x_1^2 + x_2^2 - 2 = 0,$$

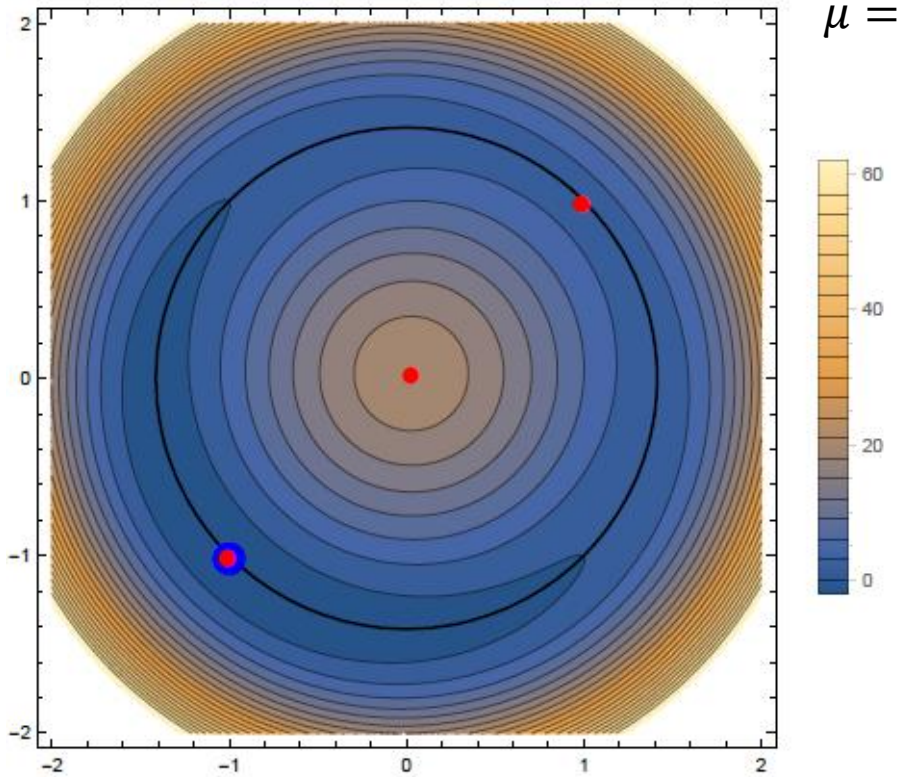
the quardratic penalty function is:

$$Q(\boldsymbol{x}, \mu) = x_1 + x_2 + \frac{\mu}{2}\left(x_1^2 + x_2^2 - 2\right)^2$$

$\mu = 1$



$\mu = 10$



Blue dot is the solution to the original problem. The constraint line is only present in the original problem. Red points are solutions to $\nabla Q = 0$.

Penalty Methods (self study)

For some problems at "small" values of μ the optimum of the penalty function may be unbounded, even if the original problem have a well-defined solution. Try increasing μ^0 if this happens.

Theorem 17.1 in Nocedal and Wright says that the global minimum of Q when $\mu \to \infty$ is also the global minimum of the original problem.

Theorem 17.2 in Nocedal and Wright says that when generating the sequence of points $\boldsymbol{x}^0, \boldsymbol{x}^1, ..., \boldsymbol{x}^k$ from minimizing Q for $\mu^0, \mu^1, ..., \mu^k \to \infty$, the limiting point $\lim_{k\to\infty} \boldsymbol{x}^k = \boldsymbol{x}^*$ may be either feasible or infeasible.

If $\boldsymbol{x}^*$ is infeasible, it is a stationary point of $\sum_{i=1}^{p} h_i^2(\boldsymbol{x})$, i.e. $\sum_{i=1}^{p} 2h_i(\boldsymbol{x}^*)\nabla h_i(\boldsymbol{x}^*) = 0$ for some $h_i(\boldsymbol{x}^*) \neq 0$. This can only happen if the gradients of $\boldsymbol{h}(\boldsymbol{x}^*)$ are linearly dependent.

If $\boldsymbol{x}^*$ is feasible and the gradients of $\boldsymbol{h}(\boldsymbol{x}^*)$ are linearly independent then theorem 17.2 says that the point $\boldsymbol{x}^*$ is a KKT point with Lagrange multipliers

$$v_i^* = -\lim_{k\to\infty} h_i(\boldsymbol{x}^k)\mu^k$$

Note that as $k \to \infty$ we have that $\mu^k \to \infty$ and $h_i(\boldsymbol{x}^k) \to 0$.

Penalty Methods (self study)

The gradient and Hessian of the penalty function are:

$$
\begin{aligned}
Q(\boldsymbol{x}, \mu) &= f(\boldsymbol{x}) + \frac{\mu}{2} \sum_{i=1}^{p} h_i^2(\boldsymbol{x}) \\
\nabla Q(\boldsymbol{x}, \mu) &= \nabla f(\boldsymbol{x}) + \mu \sum_{i=1}^{p} h_i(\boldsymbol{x}) \nabla h_i(\boldsymbol{x}) \\
\nabla^2 Q(\boldsymbol{x}, \mu) &= \nabla^2 f(\boldsymbol{x}) + \sum_{i=1}^{p} \mu h_i(\boldsymbol{x}) \nabla^2 h_i(\boldsymbol{x}) + \sum_{i=1}^{p} \mu \nabla h_i(\boldsymbol{x}) \left(\nabla h_i(\boldsymbol{x})\right)^T \\
&= \nabla^2 f(\boldsymbol{x}) + \sum_{i=1}^{p} \mu h_i(\boldsymbol{x}) \nabla^2 h_i(\boldsymbol{x}) + \mu \boldsymbol{A}^T \boldsymbol{A} \quad , \quad \boldsymbol{A}^T = \nabla \boldsymbol{h}
\end{aligned}
$$

$\boldsymbol{A}$ is a $p \times n$ matrix (the Jacobian of $\boldsymbol{h}$), and $\boldsymbol{A}^T \boldsymbol{A}$ is a $n \times n$ matrix.

When $\boldsymbol{x}$ is close to $\boldsymbol{x}^*$ (and $\boldsymbol{x}^*$ is feasible), the Hessian become

$$
\nabla^2 Q(\boldsymbol{x}, \mu \to \infty) = \nabla^2 f(\boldsymbol{x}) - \sum_{i=1}^{p} v_i^* \nabla^2 h_i(\boldsymbol{x}) + \mu \boldsymbol{A}^T \boldsymbol{A} \quad , \quad \boldsymbol{A}^T = \nabla \boldsymbol{h}
$$

where $v_i^* = -\lim_{k \to \infty} h_i(\boldsymbol{x}^k) \mu^k$ was used.

Penalty Methods (self study)

Hessian of Q:

$$\nabla^2 Q(\boldsymbol{x}, \mu \to \infty) = \nabla^2 f(\boldsymbol{x}) - \sum_{i=1}^{p} v_i^* \nabla^2 h_i(\boldsymbol{x}) + \mu \boldsymbol{A}^T \boldsymbol{A} \quad , \quad \boldsymbol{A}^T = \nabla \boldsymbol{h}$$

The first part is independent of μ and it is the Hessian of the Lagrange function of the original problem.

The second part, $\mu \boldsymbol{A}^T \boldsymbol{A}$, depend on μ and increase when μ increase.

When there are less (equality) constraints than variables, the second part is a singular matrix with its non-zero eigenvalues on the order of μ.

Thus, the condition number κ, defined as

$$\kappa\left(\nabla^2 Q\right) = \frac{\text{Largest eigenvalue}}{\text{Smallest eigenvalue}},$$

approaches ∞ as $\mu \to \infty$. The problem become more and more ill-conditioned as μ increase.

This makes the unconstrained minimization problems harder and harder to solve.

This is why we should make a sequence of solutions rather than attacking, say, $Q(\boldsymbol{x}, 10^5)$ directly. The sequence will generate good starting points for the next unconstrained minimization.

Penalty Methods

The penalty function can also be extended to handle inequality constraints.

A popular function for both equality and inequality constraints is:

$$Q(\boldsymbol{x}, \mu) = f(\boldsymbol{x}) + \frac{\mu}{2}\left(\sum_{i=1}^{p} h_i^2(\boldsymbol{x}) + \sum_{i=1}^{m} \left(\max\left(0, g_i(\boldsymbol{x})\right)\right)^2\right)$$

Note that Q above is generally not twice continuous differentiable, as we usually consider functions to be in this course!

Alternatively one could use the quardratic penalty function for equality constraint after having used slack variables to turn the inequality constraints into equality constraints.

Note: Many other penalty (and "barrier") functions are used in practice.

Augmentet Lagrangian Method also named The Method of Multipliers

The problems with the penalty method come from the ill-conditioning as μ become large, which is needed to get a highly accurate solution.

In the penalty method, the constraints are not satisfied exactly at optimum, since from Nocedal&Wright theorem 17.2:

$$h_i(\boldsymbol{x}^k) \approx -\frac{\lambda_i^*}{\mu^k}$$

The augmented Lagrangian method tries to remove the deficiencies of the penalty method by obtaining an accurate solution for a not necessarily very large value of μ.

The augmented Lagrange function is defined as:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \mu^k) = f(\boldsymbol{x}) - \sum_{i=1}^{p} v_i^k h_i(\boldsymbol{x}) + \frac{\mu^k}{2} \sum_{i=1}^{p} h_i^2(\boldsymbol{x})$$

This is a combination of the quardratic penalty function and the usual Lagrange function.

The method fixes both the Lagrange multipliers and the penalty parameter in each iteration to $\boldsymbol{v}^k$ and μ^k.

Then the problem for fixed $\boldsymbol{v}^k$ and μ^k can be solved by an unconstrained optimization method and $\boldsymbol{v}^k$ and μ^k can be updated for the next iteration.

Augmentet Lagrangian Method or the method of multipliers

The necessary conditions for the unconstrained minimization in iteration k are

$$\begin{aligned}\nabla\mathcal{L}(\boldsymbol{x}^k, \boldsymbol{v}^k, \mu^k) &= \nabla f(\boldsymbol{x}^k) - \sum_{i=1}^{p} v_i^k \nabla h_i(\boldsymbol{x}^k) + \mu^k \sum_{i=1}^{p} h_i(\boldsymbol{x}^k)\nabla h_i(\boldsymbol{x}^k) \\ &= \nabla f(\boldsymbol{x}^k) - \sum_{i=1}^{p} \left(v_i^k - \mu^k h_i(\boldsymbol{x}^k)\right) \nabla h_i(\boldsymbol{x}^k) = 0\end{aligned}$$

Compare this with the necessary conditions of the original equality-constrained problem:

$$L(\boldsymbol{x}, \boldsymbol{w}) = f(\boldsymbol{x}) - \sum_{i=1}^{p} w_i h_i(\boldsymbol{x}) \quad \Rightarrow \quad \nabla L(\boldsymbol{x}^*, \boldsymbol{w}^*) = \nabla f(\boldsymbol{x}^*) - \sum_{i=1}^{p} w_i^* \nabla h_i(\boldsymbol{x}^*) = 0$$

If $v_i^k - \mu^k h_i(\boldsymbol{x}^k) = w_i^*$, these conditions are the same. Thus, by choosing the Lagrange multipliers v_i^k such that

$$v_i^k - \mu^k h_i(\boldsymbol{x}^k) = w_i^* \quad \Rightarrow \quad h_i(\boldsymbol{x}^k) = \frac{v_i^k - w_i^*}{\mu^k}$$

we can obtain a solution that satisfies the equality constraints accurately and that satisfies the necessary conditions of the original constrained problem, but without necessarily have a large μ^k.

When v_i^k is close to w_i^*, the constraints are satisfied much better than in the Penalty method, for the same μ^k.

Augmentet Lagrangian Method or the method of multipliers

$w_i^* = v_i^k - \mu^k h_i(\boldsymbol{x}^k)$ suggest the following method for updating the Lagrange multipliers:

$$v_i^{k+1} = v_i^k - \mu^k h_i(\boldsymbol{x}^k)$$

Augmented Lagrangian Method:

Choose some starting parameters, μ^0, $\boldsymbol{x}^{-1}$, and $\boldsymbol{v}^0$.
Choose a starting tolerance, ϵ^0, and stopping criteria, τ and μ_{max}.
Set $k = 0$.
While $\mu^k < \mu_{max}$ and ($k = 0$ or $||\boldsymbol{x}^{k-1} - \boldsymbol{x}^{k-2}|| > \tau$)
    Find $\boldsymbol{x}^k$ by
      minimizing $\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \mu^k)$ wrt. $\boldsymbol{x}$ from starting point $\boldsymbol{x}^{k-1}$ to tolerence ϵ^k.
    $v_i^{k+1} = v_i^k - \mu^k h_i(\boldsymbol{x}^k)$.
    Choose $\mu^{k+1} > \mu^k$ and $\epsilon^{k+1} \leq \epsilon^k$.
    $k = k + 1$.

Often $\mu^{k+1} = K_\mu \mu^k$ and $\epsilon^{k+1} = \epsilon^k / K_\epsilon$ is used for some constants $K_\mu > 1$ and $K_\epsilon \geq 1$.

We often get away with a (much) smaller K_μ than the one used in the quadratic penalty method, but an initial too small K_μ may lead to an unbounded problem.

Next time…

We will apply the augmented Lagrangian method to handle the many stress constraints.